

Complex Transseries Reversion at q -Cusps

Convergent normal forms, infinitely flat limits,
logarithmic–Puisseux inverses, and recovery from cusp data

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Abstract

We develop a branch-resolved inversion calculus for complex transseries and apply it to q -products at rational boundary points. The analytic core is a convergent, multivariable implicit-function construction for $Y = Ct^\beta e^{-A/t} U(t, e^{-B_1/t}, \dots, e^{-B_d/t})$. It retains complete exponential sectors, gives explicit coefficient extraction and residual bounds, and converts ratios of actions into powers of the inverse variable. A second scaling handles power and regular cores whose exponentially small perturbations have large logarithmic derivatives.

For every finite Euler-product quotient $\prod_{\delta|N} (q^\delta; q^\delta)_\infty^{r_\delta}$, we derive an exact analytic normal form at every root of unity. Three explicit linear invariants determine the inverse regime. When all three vanish, the quotient has a nonconstant but infinitely flat finite limit; its inverse is described by a convergent Puiseux series inside a logarithmic denominator. We prove a four-factor minimum at $q = 1$ and realize every finite correction ramification degree with four factors. We compute two examples and recover every exponent r_δ from the leading exponential data at divisor cusps by a double Möbius inversion. Additional results treat nonmodular shifted factorials, finite Gaussian multinomials, and the transport of sectorial jumps through inversion.

All stated local theorems have conventional proofs. Classical modular, Lagrange, and resurgent-closure results are attributed; the proposed contribution is their explicit branch-aware synthesis and the accompanying classification and reconstruction formulas. Worldwide novelty is not established. The included computations are reproducible checks, not proof-assistant or interval certificates.

Self-contained source and bibliography.

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1 Scope, source boundary, and the questions answered

1.1 What “general complex transseries” can responsibly mean

Series reversal means compositional inversion, not reversal of the order of coefficients. Over the complex plane, the choice of sector, logarithm, ramification, and analytic realization is part of the inversion problem. There cannot be a single branch-free inverse calculus even for $Y = e^{-1/t}$: its inverses are $-1/(\text{Log } Y + 2\pi i k)$, $k \in \mathbb{Z}$.

We distinguish three levels throughout. A *formal* statement takes place in a specified complete filtered algebra. A *sectorial* statement concerns actual holomorphic functions with uniform estimates on smaller sectors. A *global continuation* statement requires transition maps and their domains. No implication from the first level to the third is assumed. In particular, attaching complex coefficients to a real transseries field does not by itself define all oscillatory compositions or all analytic sums.

The generality proved here is a preparation principle: after a dominant phase is inverted and the remaining small quantities are promoted to independent parameters, a substantial class of complex transseries becomes a convergent analytic implicit-function problem. The formal version also works in other complete filtered algebras when its contraction hypotheses are verified. We do not assert that every conceivable Hahn support, accumulating action set, or divergent transseries admits such a preparation.

1.2 Relation to ProveIt and to existing literature

The inspected repository snapshot is the commit printed on the title page. The canonical transseries directory records a large inversion volume, a combinatorial companion, and separate research arrivals; it explicitly distinguishes human arguments from exact Lean counterparts [1]. The companion already contains phase-coordinate reversal, analytic exponential-tail inversion, and applications to q -products. Consequently, neither “use Lagrange inversion” nor “invert an exponential tail” is presented here as a new foundational discovery.

The inspected *Certified Inversion Through the $q \rightarrow 1$ Transition* README describes signed remainders, Stieltjes–Padé bounds, optimal truncation, and a cancellation window for Gaussian products [2]. The *Beyond Finite-Action Folds* README concerns countable-action critical models and Hahn inverses [3]. Those are different problems from classifying an Euler quotient’s finite flat limit at every rational cusp. This is a targeted crosswalk based on the listed documentation, not a claim to have audited all pages of the canonical volume or every incoming article.

Formal and analytic nonlinear operations on resurgent functions, including functional and implicit inversion, are established subjects. Sauzin proves precise closure and summability theorems in an appropriate resurgent setting [4]. We use those results as background, not as an open conjecture supposedly solved here. The modular transformation of Dedekind’s eta function is classical [5]. All-order q -factorial asymptotics involving polylogarithms and Bernoulli polynomials occur in McIntosh’s work [7], and root-of-unity asymptotics are central to the work of Garoufalidis and Zagier [8].

Two recent sources materially sharpen the source boundary. Fantini and Rella analyze resurgence and summability of q -Pochhammer symbols and weighted combinations [9]. Arabi Ardehali and Rosengren give a gamma-product representation and asymptotic applications, with a version of record published on 19 September 2026 [10]. We do not claim a new q -Pochhammer product formula or a solution of their general summability questions. Our exact results use the modular Euler-product subclass; Section 10 separately treats an asymptotic,

generally nonconvergent, shifted-factorial example.

1.3 A precise result ledger

The following concrete questions are answered in this article.

- Q1.** Can inversion preserve full exponential sectors rather than just algebraic coefficients? Theorem 3.1 gives a convergent multivariable lift, sector-preserving truncation, and action transport.
- Q2.** What replaces ordinary Puiseux inversion at a nonconstant finite limit invisible to every power of t ? Theorem 5.1 gives all local sheets, their monodromy, and an explicit error law.
- Q3.** Which finite Euler quotients have such a limit at a specified root of unity? Theorem 7.1 gives a necessary-and-sufficient three-invariant test and an inverse in each complementary regime.
- Q4.** How few factors can produce flatness at $q = 1$? Four distinct scales are necessary and sufficient, with an explicit kernel formula (Theorem 7.2).
- Q5.** Can all rational cusps conceal the quotient's exponent vector? No: Theorem 9.1 gives exact recovery from divisor-cusp exponential data.
- Q6.** Is finite correction ramification bounded by the number of product factors? No: four factors realize every positive degree, with an exact intrinsic-degree criterion (Theorem 7.3).

These are explicitly posed local questions. Answering them is not a claim of resolving an independently documented major open problem. The proofs and formulations are research contributions of this draft; publication priority and independent validation remain separate matters.

2 Sectorial preparation and reliable inverse certificates

2.1 Small actions and admissible sectors

A typical asymptotic sector is

$$S(\theta, \alpha, r) = \{0 < |t| < r : |\arg t - \theta| < \alpha\},$$

with its argument understood on a chosen logarithmic cover. Bounds are uniform on closed, strictly smaller subsectors. A quantity $e^{-B/t}$ is small only where $\Re(B/t) > 0$; it is uniformly exponentially small where $\Re(B/t) \geq c/|t|$ for some $c > 0$.

For finitely many actions λ_j , assume a sector of large x on which $\Re(\lambda_j x) \geq c|x|$ for every j . Their additive semigroup is locally finite when measured by this positive real projection: only finitely many multi-indices have projection at most a given bound. This permits resonant actions to be collected without infinite coefficient multiplicities. It does not supply a total asymptotic ordering on every direction. In particular, a dominance wall is not automatically a Stokes jump.

Lemma 2.1 (Complete filtered contraction). *Let $\mathcal{A}$ be a complete separated $\mathbb{C}$ -algebra with descending multiplicative ideals I^n , $n \geq 0$. Suppose a map $T : I \rightarrow I$ is defined by admissible*

algebraic or Taylor operations and

$$u - v \in I^n \implies T(u) - T(v) \in I^{n+1}.$$

Then $u = T(u)$ has a unique solution in I . Its residue modulo I^{n+1} is obtained after n Picard steps from 0.

Proof. Put $u_0 = 0$ and $u_{n+1} = T(u_n)$. Induction gives $u_{n+1} - u_n \in I^{n+1}$. Completeness gives a limit u ; the displayed condition implies continuity and hence $u = T(u)$. For two fixed points, $u - v \in I$ improves successively to every I^n , so separatedness gives $u = v$. The same induction gives the residue assertion. $\square$

The admissibility clause is essential. One must not differentiate, compose, or sum arbitrary Hahn families merely because the resulting expression is suggestive. For analytic parameter germs we use $\mathbb{C}[[v, Q_1, \dots, Q_d]]$ and its maximal-ideal filtration; all operations below are then explicit and admissible.

2.2 A complex residual certificate with existence

Lemma 2.2 (Derivative-close-to-one certificate). *Let K be holomorphic on a neighborhood of the closed disk $\overline{D}(z_*, r)$. Suppose*

$$\sup_{\overline{D}(z_*, r)} |K'(z) - 1| \leq \kappa < 1, \quad |K(z_*)| < (1 - \kappa)r.$$

Then K has a unique zero z in that disk and

$$|z - z_*| \leq \frac{|K(z_*)|}{1 - \kappa}. \quad (2.1)$$

Proof. The map $T(w) = w - K(w)$ has Lipschitz constant at most κ on the convex disk. Moreover, $|T(w) - z_*| \leq \kappa|w - z_*| + |K(z_*)| < r$. The contraction theorem gives a unique fixed point. At that point, $|z - z_*| \leq \kappa|z - z_*| + |K(z_*)|$, proving the bound. $\square$

A lower bound on $|K'|$ alone is not a substitute for this argument in the complex plane: it does not prove injectivity on an arbitrary region. The lemma supplies both existence and the backward-to-forward error conversion needed by an inverse algorithm.

3 A convergent lift for prepared exponential transseries

3.1 The exact normal form

Fix $A, C \neq 0$, $\beta \in \mathbb{C}$, and actions $B_1, \dots, B_d \in \mathbb{C}$. Let $U(t, \mathbf{E})$ be holomorphic near $(0, \mathbf{0})$ with $U(0, \mathbf{0}) = 1$. Consider

$$Y = Ct^\beta e^{-A/t} U(t, e^{-B_1/t}, \dots, e^{-B_d/t}). \quad (3.1)$$

Choose all logarithms coherently so that, with $x = A/t$, $\text{Log } t = \text{Log } A - \text{Log } x$. Put

$$L = \text{Log } \frac{CA^\beta}{Y}, \quad x_0 + \beta \text{Log } x_0 = L, \quad v = x_0^{-1}, \quad \lambda_j = B_j/A, \quad Q_j = e^{-\lambda_j x_0}. \quad (3.2)$$

On a closed large- L sector away from the logarithmic cut, the root with $x_0 \sim L$ exists and is unique in an $O(\log |L|)$ disk around L . Indeed, $x \mapsto L - \beta \operatorname{Log} x$ maps such a disk to itself and has Lipschitz constant $O(|L|^{-1})$. Thus

$$x_0 = L - \beta \operatorname{Log} L + O(\log |L|/|L|). \quad (3.3)$$

When $\beta \neq 0$, the identity $x_0 = \beta W(e^{L/\beta}/\beta)$ is available, but the branch is the one selected by (3.2); it need not be the principal W branch.

Theorem 3.1 (Analytic parameter lift). *There is a unique holomorphic germ $D(v, \mathbf{Q})$ at the origin, with $D(0, \mathbf{0}) = 0$, satisfying*

$$\begin{aligned} D = & -\beta \log(1 + vD) \\ & + \log U \left(\frac{Av}{1 + vD}, Q_1 e^{-\lambda_1 D}, \dots, Q_d e^{-\lambda_d D} \right). \end{aligned} \quad (3.4)$$

In every sector in which $|x_0| \rightarrow \infty$ and $\Re(\lambda_j x_0) \geq c|x_0|$, the inverse of (3.1) near A/x_0 is exactly

$$t = \frac{A}{x_0 + D(x_0^{-1}, e^{-\lambda_1 x_0}, \dots, e^{-\lambda_d x_0})}. \quad (3.5)$$

Its multivariable Taylor expansion is convergent, and no exponential action outside the additive semigroup generated by the λ_j is needed in the x_0 coordinate.

Proof. Subtract the logarithm of (3.1) from (3.2) and write $x = x_0 + D$. The resulting equation is (3.4). Its left side minus right side is holomorphic in $(D, v, \mathbf{Q})$ near zero. Its derivative with respect to D there is 1: the logarithmic term has a factor v , the first argument of U has a factor v , and each other argument has a factor Q_j . The analytic implicit-function theorem gives the germ and uniqueness. The right side also has positive parameter order and raises the order of a difference in D , so Lemma 2.1 gives the same formal germ.

Under the sector hypotheses the actual parameters tend to the origin. Substitution into the convergent germ solves the original equation with its chosen logarithms. The prepared equation has derivative close to one on a fixed small D disk, giving local uniqueness by Lemma 2.2. Finally, a Taylor monomial $v^k \mathbf{Q}^n$ evaluates to $x_0^{-k} e^{-(n \cdot \boldsymbol{\lambda})x_0}$. Convergence and the positive projection condition justify grouping resonant actions. Analytic reciprocation in (3.5) preserves the same action semigroup. $\square$

3.2 Coefficient extraction and sector-preserving truncation

Write the right side of (3.4), with D replaced by z , as $\Phi(z; v, \mathbf{Q})$. Classical Lagrange inversion with an auxiliary parameter gives

$$D = \sum_{n \geq 1} \frac{1}{n!} \frac{\partial^{n-1}}{\partial z^{n-1}} \Phi(z; v, \mathbf{Q})^n \Big|_{z=0}. \quad (3.6)$$

This is a coefficientwise identity in the parameter filtration: every summand has order at least n . It is also an analytic identity for sufficiently small parameters. To see the latter directly, on a fixed small z disk the whole Φ is uniformly small, so the auxiliary parameter can vary in a disk of radius larger than 1.

Suppose D is bounded by M on a closed polydisk of radius R , slightly inside its domain. If $D_{\leq n}$ denotes total-parameter-degree truncation and $s = (|v| + \sum_j |Q_j|)/R < 1$, Cauchy's estimate yields

$$|D - D_{\leq n}| \leq M \frac{s^{n+1}}{1 - s}. \quad (3.7)$$

The sum of monomials of degree k is bounded by the k th power of the sum of their absolute variables; no missing multinomial factor is needed.

For exponential accuracy, a different grouping is preferable:

$$D(v, \mathbf{Q}) = \sum_{\mathbf{n} \in \mathbb{N}^d} D_{\mathbf{n}}(v) \mathbf{Q}^{\mathbf{n}}. \quad (3.8)$$

Keep the coefficient functions $D_{\mathbf{n}}(v)$ intact. Uniformly for v in a smaller disk, the tail after $|\mathbf{n}| \leq n$ is bounded by the analogue of (3.7) with $s = \sum_j |Q_j|/R$. Truncating every coefficient function to a fixed algebraic order instead leaves an error much larger than an exponential sector. Thus a finite algebraic approximation does not automatically certify a beyond-all-orders claim.

For an approximate displacement $\tilde{D}$, apply Lemma 2.2 to the left side minus right side of (3.4). Once both displacements are in its disk,

$$|t - \tilde{t}| \leq \frac{|A| |K(\tilde{D})|}{(1 - \kappa) |x_0 + D| |x_0 + \tilde{D}|}. \quad (3.9)$$

This is an explicit certificate, including a root-existence test.

3.3 Actions become inverse-variable exponents

The core identity gives the exact branch-dependent relation

$$Q_j = e^{-\lambda_j L} x_0^{\beta \lambda_j} = \left(\frac{Y}{C A^\beta} \right)^{\lambda_j} x_0^{\beta \lambda_j}. \quad (3.10)$$

Thus B_j/A , not B_j alone, is the exponent visible in the inverse variable. Logarithmic dressing comes from the core x_0 . All powers in this formula are defined through the same logarithm as L .

Proposition 3.2 (A sharp one-action ramification test). *For $a, b > 0$, let $Y = ce^{-a/t}(1 + e^{-b/t})$, $c \neq 0$, and set $\lambda = b/a$, $L = \text{Log}(c/Y)$. Its inverse is $t = a/(L + D((Y/c)^\lambda))$, where D is convergent near zero and*

$$D(Q) = Q - (\lambda + \tfrac{1}{2})Q^2 + O(Q^3).$$

The correction to the inverse phase is a convergent Puiseux germ in Y/c after finite ramification if and only if $\lambda \in \mathbb{Q}$.

Proof. The implicit equation is $D = \log(1 + Qe^{-\lambda D})$. It gives the displayed coefficients and $D'(0) = 1$. Rational λ clearly suffices. Conversely, a nonzero convergent Puiseux germ has a rational first exponent. The first exponent here is exactly λ , so irrational λ is impossible. The logarithm in L is retained; the assertion concerns its analytic phase correction, not removal of logarithmic monodromy. $\square$

4 Flat perturbations of power and regular cores

When the leading essential action vanishes, Theorem 3.1 is no longer the right coordinate chart. Consider instead

$$Y = Ct^\beta e^{Bt} H(e^{-a/t}), \quad a > 0, \quad H(0) = 1, \quad (4.1)$$

with H holomorphic near zero. Assume $(\beta, B) \neq (0, 0)$. Let τ be a chosen small root of the *exact core equation*

$$Y = C\tau^\beta e^{B\tau}. \quad (4.2)$$

For $\beta \neq 0$, choose a branch of $z = (Y/C)^{1/\beta}$; then $\tau = \beta W((B/\beta)z)/B$ near zero when $B \neq 0$, and $\tau = z$ when $B = 0$. For $\beta = 0$, $B \neq 0$, $\tau = B^{-1} \text{Log}(Y/C)$.

Theorem 4.1 (Slope-adapted analytic lift). *Put $\sigma = 1$ if $\beta \neq 0$, and $\sigma = 2$ if $\beta = 0$, $B \neq 0$. There is a holomorphic germ $D(\tau, R)$ with $D(\tau, 0) = 0$ for which the exact local inverse is*

$$t = \tau + \tau^2 D\left(\tau, \frac{e^{-a/\tau}}{\tau^\sigma}\right). \quad (4.3)$$

This is a convergent transseries on closed subsectors of $\Re(1/\tau) > 0$. If $H(Q) = 1 + h_1 Q + O(Q^2)$, its first displacement is

$$t - \tau \sim \begin{cases} -\frac{h_1}{\beta} \tau e^{-a/\tau}, & \beta \neq 0, \\ -\frac{h_1}{B} e^{-a/\tau}, & \beta = 0, B \neq 0, \end{cases} \quad (4.4)$$

provided $h_1 \neq 0$.

Proof. Write $t = \tau(1 + \tau D)$. Then

$$e^{-a/t} = e^{-a/\tau} \exp\left(\frac{aD}{1 + \tau D}\right).$$

Subtract the logarithm of the core equation. For $\beta \neq 0$, divide by τ to obtain

$$0 = \beta \frac{\log(1 + \tau D)}{\tau} + B\tau D + \frac{1}{\tau} \log H\left(\tau R \exp\left(\frac{aD}{1 + \tau D}\right)\right).$$

All apparent singularities at $\tau = 0$ are removable. The derivative with respect to D at $(D, \tau, R) = 0$ is $\beta \neq 0$. For $\beta = 0$, divide instead by τ^2 :

$$0 = BD + \frac{1}{\tau^2} \log H\left(\tau^2 R \exp\left(\frac{aD}{1 + \tau D}\right)\right).$$

The derivative is now $B \neq 0$. The analytic implicit-function theorem proves both cases; setting $R = 0$ gives $D = 0$. Exponential decay ensures $e^{-a/\tau}/\tau^\sigma \rightarrow 0$ in the stated subsectors. Linearization in R , followed by $\tau \rightarrow 0$, gives (4.4). $\square$

The division by τ or τ^2 is not cosmetic. It absorbs the large logarithmic derivative of the exponential into the small parameter. Expanding only in Q while pretending that all coefficients stay uniformly bounded at $\tau = 0$ would miss this scaling.

5 Infinitely flat finite limits and logarithmic–Puisseux inversion

5.1 All sheets and a convergent denominator

Suppose

$$Y(t) = Y_* + ce^{-ma/t}U(e^{-a/t}), \quad a > 0, \quad c \neq 0, \quad m \in \mathbb{N}_{>0}, \quad U(0) = 1, \quad (5.1)$$

where U is holomorphic near zero. Let s and $\log s$ be fixed on a simply connected punctured neighborhood by

$$s = \left(\frac{Y - Y_*}{c} \right)^{1/m}, \quad \log s = \frac{1}{m} \operatorname{Log} \frac{Y - Y_*}{c}.$$

Theorem 5.1 (Logarithmic–Puiseux inverse atlas). *Let ψ be the analytic inverse at zero of $f(Q) = QU(Q)^{1/m}$, where $U^{1/m}(0) = 1$, and let $V(z) = \log(\psi(z)/z)$ with $V(0) = 0$. Then all solutions with $|e^{-a/t}|$ sufficiently small are*

$$t_\ell(Y) = -\frac{a}{\Lambda_\ell + V(s_\ell)}, \quad s_\ell = e^{2\pi i \ell/m} s, \quad \Lambda_\ell = \log s + \frac{2\pi i \ell}{m}, \quad \ell \in \mathbb{Z}. \quad (5.2)$$

The coefficients are explicit:

$$\psi(z) = \sum_{n \geq 1} \frac{z^n}{n} [Q^{n-1}] U(Q)^{-n/m}. \quad (5.3)$$

A positive circuit of Y around Y_* continues sheet ℓ to sheet $\ell + 1$.

Proof. The map f is holomorphic, $f(0) = 0$, and $f'(0) = 1$. Hence it has a unique local analytic inverse ψ . With $E = e^{-a/t}$, (5.1) becomes $f(E)^m = s^m$, so $E = \psi(e^{2\pi i j/m} s)$ for one j modulo m . The logarithms of this nonzero E are

$$\log s + 2\pi i j/m + V(e^{2\pi i j/m} s) + 2\pi i k, \quad k \in \mathbb{Z}.$$

Writing $\ell = j + mk$ gives exactly (5.2), with no independent or double-counted branch labels. Conversely, exponentiating that equation recovers E and then Y . Formula (5.3) is ordinary Lagrange inversion. Under a positive circuit of $Y - Y_*$, s continues to $e^{2\pi i/m} s$ and $\log s$ increases by $2\pi i/m$; substitution proves the monodromy assertion. $\square$

Although the cusp lies on the boundary of the original q disk, the inverse branches are actual holomorphic functions on slit target neighborhoods. For small s , $\Re(\Lambda_\ell + V(s_\ell)) < 0$; therefore $\Re t_\ell > 0$. For each fixed ℓ , $t_\ell \rightarrow 0$ as $Y \rightarrow Y_*$. The exact formula also permits indices that grow, but claims about uniform sectorial asymptotics then require a separate bound on the argument of the denominator.

5.2 Remainder law and sensitivity

Corollary 5.2 (Explicit inverse remainder). *Suppose V is holomorphic near the closed disk $|z| \leq R$, $|V| \leq M$ there with $M > 0$, and V_N is its Taylor polynomial through degree N . For $\rho = |s|/R \leq 1/2$ and $|\Lambda_\ell| \geq 2M$,*

$$\left| t_\ell + \frac{a}{\Lambda_\ell + V_N(s_\ell)} \right| \leq \frac{4aM}{|\Lambda_\ell|^2} \frac{\rho^{N+1}}{1 - \rho}. \quad (5.4)$$

For fixed sheet, this is $O(|Y - Y_*|^{(N+1)/m} / |\operatorname{Log}(Y - Y_*)|^2)$.

Proof. Cauchy's bound gives $|V - V_N| \leq M\rho^{N+1}/(1 - \rho)$. Because $V(0) = 0$, $|V_N| \leq M \sum_{j=1}^N \rho^j \leq M$; also $|V| \leq M$. Both denominators have modulus at least $|\Lambda_\ell|/2$. Subtract the two reciprocals to get (5.4). $\square$

Differentiating (5.1), with $E = e^{-a/t}$, gives

$$\frac{dt}{dY} = \frac{t^2}{ma(Y - Y_*)}(1 + O(E)). \quad (5.5)$$

Thus the inverse is badly conditioned with respect to *absolute* errors in Y near Y_* . In relative displacement units, however, $(Y - Y_*)t'(Y)/t$ is only of order $1/|\text{Log}(Y - Y_*)|$ on a fixed sheet. An algorithm should therefore compute $Y - Y_*$ directly, or with enough precision to survive its cancellation, rather than subtracting two low-precision numbers close to Y_* .

Every map in (5.1) has the same algebraic asymptotic expansion $Y_* + 0t + 0t^2 + \dots$, irrespective of a, m, c, U . Consequently, algebraic coefficients alone do not determine the inverse, its leading scale, or its branch structure. The full transseries data do. This is a failure of information loss under projection to a power series, not a failure of transseries inversion.

6 An exact normal form at every rational q -cusp

6.1 Euler quotients and cusp coordinates

Write

$$\mathcal{P}(q) = (q; q)_\infty = \prod_{n \geq 1} (1 - q^n), \quad |q| < 1.$$

Let $N \geq 1$ and let $r_\delta \in \mathbb{Z}$, $\delta \mid N$, not all vanish. Our main family is

$$\mathcal{F}_r(q) = \prod_{\delta \mid N} \mathcal{P}(q^\delta)^{r_\delta}. \quad (6.1)$$

Zero exponents may be suppressed. Each factor is nonzero in the unit disk, so negative exponents cause no poles there. We call (6.1) an *Euler quotient*: converting it to a literal eta quotient introduces a known elementary power of q . No modular-form congruences on r are imposed.

Choose coprime integers h, k , $k > 0$, and put

$$\zeta = e^{2\pi i h/k}, \quad q = \zeta e^{-t}, \quad \Re t > 0.$$

The cusp $q = 1$ is represented by $(h, k) = (0, 1)$. For each active scale set

$$g_\delta = \gcd(k, \delta), \quad c_\delta = k/g_\delta, \quad a_\delta = \delta h/g_\delta. \quad (6.2)$$

Choose d_δ with $a_\delta d_\delta \equiv 1 \pmod{c_\delta}$; when $c_\delta = 1$, take $d_\delta = 0$. Define

$$s(d, c) = \sum_{n=1}^{c-1} \left(\frac{n}{c} - \frac{1}{2} \right) \left(\frac{dn \bmod c}{c} - \frac{1}{2} \right), \quad s(d, 1) = 0. \quad (6.3)$$

For $c > 1$ the residues in this sum are nonzero because d is coprime to c . Thus this is the usual Dedekind sum.

The data needed for inversion are

$$R_0 = \sum_{\delta \mid N} r_\delta, \quad R_1 = \sum_{\delta \mid N} \delta r_\delta, \quad \beta = -R_0/2, \quad B = R_1/24, \quad (6.4)$$

$$A_k = \frac{\pi^2}{6k^2} \sum_{\delta \mid N} \frac{r_\delta g_\delta^2}{\delta}, \quad a = \frac{4\pi^2}{k^2 N}, \quad m_\delta = \frac{N g_\delta^2}{\delta}, \quad \rho_\delta = e^{-2\pi i d_\delta / c_\delta}. \quad (6.5)$$

Every m_δ is a positive integer. Finally put

$$C_{h,k} = (2\pi)^{R_0/2} \prod_{\delta|N} (c_\delta \delta)^{-r_\delta/2} \exp \left(-\pi i \sum_{\delta|N} r_\delta s(d_\delta, c_\delta) \right), \quad (6.6)$$

$$H_{h,k}(Q) = \prod_{\delta|N} \mathcal{P}(\rho_\delta Q^{m_\delta})^{r_\delta}. \quad (6.7)$$

The function $H_{h,k}$ is holomorphic and nonzero for $|Q| < 1$, and $H_{h,k}(0) = 1$. The number A_k depends only on the cusp denominator; $C_{h,k}$ and the phases in $H_{h,k}$ also depend on the numerator.

Theorem 6.1 (Exact cusp preparation). *For every t with $\Re t > 0$,*

$$\mathcal{F}_r(\zeta e^{-t}) = C_{h,k} t^\beta \exp(-A_k/t + Bt) H_{h,k}(e^{-a/t}). \quad (6.8)$$

Here t^β uses the logarithm on the right half-plane that is real on the positive axis. This is an identity of holomorphic functions, not merely an asymptotic expansion.

Proof. We record the reduction to the classical eta transformation [5, §27.14] to fix all phases. For one scale omit the subscript δ on $a_\delta, c_\delta, d_\delta$, and choose an integer b with $ad - bc = 1$. Set

$$z = \frac{a}{c} + \frac{i\delta t}{2\pi}, \quad w = \frac{dz - b}{-cz + a} = -\frac{d}{c} + \frac{2\pi i}{c^2 \delta t}.$$

Both z, w lie in the upper half-plane. In a compatible square-root convention the eta law is

$$\eta \left(\frac{aw + b}{cw + d} \right) = \exp \left(\pi i \left(\frac{a+d}{12c} - s(d, c) \right) \right) [-i(cw + d)]^{1/2} \eta(w).$$

Use $\mathcal{P}(e^{2\pi iz}) = e^{-\pi iz/12} \eta(z)$ and $cw + d = 2\pi i/(c\delta t)$. The result is

$$\begin{aligned} \mathcal{P}(\zeta^\delta e^{-\delta t}) &= e^{-\pi i s(d, c)} \left(\frac{2\pi}{c\delta t} \right)^{1/2} \exp \left(-\frac{\pi^2}{6c^2 \delta t} + \frac{\delta t}{24} \right) \\ &\quad \cdot \mathcal{P} \left(e^{-2\pi i d/c} e^{-4\pi^2/(c^2 \delta t)} \right). \end{aligned} \quad (6.9)$$

In particular, the terms $(a+d)/(12c)$ cancel exactly; discarding them before transforming would give the wrong phase. The square root in (6.9) is positive for positive t before its displayed Dedekind multiplier. Multiplying the individual formulas to powers r_δ gives (6.8) and all its constants. The formula for $c = 1$ includes integer translations of z and has the same cancellation. No modular-form transformation property for the assembled quotient was needed. $\square$

6.2 Exact coefficients and termination of the first-action search

Let $\sigma_1(n) = \sum_{d|n} d$. Writing $\log H_{h,k}(Q) = \sum_{n \geq 1} \ell_n Q^n$, direct expansion of $\log \mathcal{P}(z)$ gives

$$\ell_n = -\frac{1}{n} \sum_{\substack{\delta|N \\ m_\delta|n}} r_\delta m_\delta \sigma_1(n/m_\delta) \rho_\delta^{n/m_\delta}. \quad (6.10)$$

If $H_{h,k}(Q) = \sum_{n \geq 0} h_n Q^n$, then

$$h_0 = 1, \quad nh_n = \sum_{j=1}^n j \ell_j h_{n-j} \quad (n \geq 1). \quad (6.11)$$

Thus every coefficient is in an explicitly specified cyclotomic field. At $q = 1$, all phases are 1 and the coefficients are integers: this also follows directly from integer powers of the Euler factors.

Lemma 6.2 (The transformed tail is nonconstant). *For a nonzero finite exponent vector $\mathbf{r}$, the function $H_{h,k}$ is not identically 1. Consequently the search for the first nonzero coefficient of $H_{h,k} - 1$ terminates at every cusp.*

Proof. Suppose $H_{h,k} = 1$. By analytic continuation in the unit disk it is identically 1 throughout its domain. Formula (6.8) then holds without a tail for every real $t > 0$. As $t \rightarrow +\infty$, $\mathcal{F}_{\mathbf{r}}(\zeta e^{-t}) = 1 + O(e^{-\delta_0 t})$, where δ_0 is the smallest active scale. Since B, β, A_k are real, comparison with $C_{h,k} t^\beta e^{-A_k/t + Bt}$ first gives $B = 0$, then $\beta = 0$ and $C_{h,k} = 1$, and finally $A_k = 0$ by comparison of the $1/t$ terms. It follows that $\mathcal{F}_{\mathbf{r}} = 1$. But its Taylor coefficient at q^{δ_0} is $-r_{\delta_0} \neq 0$; larger scales cannot contribute to that coefficient. This contradiction proves the claim. $\square$

The proof is qualitative: it does not give a useful uniform bound for the first nonzero index at arbitrary cusps. At $q = 1$ there is a simple bound and an exact answer. If $\delta_{\max}$ is the largest active scale, the smallest $m_\delta = N/\delta$ is unique, so

$$m = \text{ord}_0(H_{0,1} - 1) = N/\delta_{\max}, \quad h_m = -r_{\delta_{\max}}. \quad (6.12)$$

Changing the auxiliary common multiple N merely changes the coordinate Q ; the physical first action $ma = 4\pi^2/\delta_{\max}$ is unchanged. An apparent Puiseux order can contain redundant ramification if H is a function of a proper power of Q . The complete sheet formula remains valid; a smaller primitive action coordinate removes that redundancy.

7 Classification of cusp inverses and a four-factor minimum

Theorem 7.1 (Four inverse regimes). *For a nontrivial Euler quotient (6.1) at a fixed cusp, exactly one of the following applies, with the data of Theorem 6.1.*

- (I) $A_k \neq 0$. Along positive t , the quotient tends to 0 when $A_k > 0$ and has unbounded modulus when $A_k < 0$. Its inverse has a logarithmic phase core and a convergent analytic lift as in Theorem 3.1.
- (II) $A_k = 0$ and $\beta \neq 0$. The inverse has a power core, perturbed by a convergent slope-adapted exponential transseries as in Theorem 4.1.
- (III) $A_k = \beta = 0$ and $B \neq 0$. The finite limit is $C_{h,k}$. The exact regular core is $\tau = B^{-1} \text{Log}(Y/C_{h,k})$, and its flat perturbation is given by Theorem 4.1 with $\sigma = 2$.
- (IV) $A_k = \beta = B = 0$. The finite limit is $C_{h,k}$ and is infinitely flat on closed right-half-plane subsectors. Put $m = \text{ord}_0(H_{h,k} - 1)$ and $c = C_{h,k} h_m$. Then all local inverse sheets are given by Theorem 5.1, with

$$U(Q) = \frac{H_{h,k}(Q) - 1}{h_m Q^m}.$$

In particular, a nonconstant infinitely flat finite nonzero limit occurs if and only if the three linear conditions

$$\sum_{\delta|N} \frac{r_\delta \gcd(k, \delta)^2}{\delta} = 0, \quad \sum_{\delta|N} r_\delta = 0, \quad \sum_{\delta|N} \delta r_\delta = 0 \quad (7.1)$$

hold.

Proof. The alternatives are exhaustive and disjoint by their definitions. For (I), put $\epsilon = \text{sign}(A_k)$ and invert $Y = \mathcal{F}_r^\epsilon$ instead of $\mathcal{F}_r$ when necessary. The prepared data are $A = \epsilon A_k > 0$, $C = C_{h,k}^\epsilon$, $\beta' = \epsilon\beta$ and $U(t, E) = e^{\epsilon Bt} H_{h,k}(E)^\epsilon$, with single action $a > 0$. The reciprocal target change is locally invertible and restores the original target at the end. Theorem 3.1 applies. Alternatives (II) and (III) are exactly (4.1). For (IV), Lemma 6.2 gives a finite m and nonzero h_m ; its displayed U is a holomorphic unit. This proves applicability of Theorem 5.1.

For necessity of the final assertion, compare (6.8) along $t \downarrow 0$. A finite nonzero limit forces $A_k = 0$ and $\beta = 0$, since both are real. The remaining algebraic factor is e^{Bt} . Flatness forces its linear coefficient B to vanish. Conversely, all three vanishing leaves a nonconstant exponentially small tail. This is flat uniformly wherever $\Re(1/t) \geq c/|t|$. $\square$

The original inverse in the q plane is simply $q(Y) = \zeta \exp(-t(Y))$. Analytic composition preserves the local sheet information and the proved error estimates, up to a bounded derivative factor on each sufficiently small chart. We give formulas in t because it is the natural cusp coordinate and displays the transseries scales.

7.1 Minimum support and all four-scale solutions at $q = 1$

At $q = 1$, the three conditions are

$$\sum_j r_j / \delta_j = 0, \quad \sum_j r_j = 0, \quad \sum_j \delta_j r_j = 0. \quad (7.2)$$

Theorem 7.2 (Four-factor minimum). *On s distinct positive integer scales $\delta_1, \dots, \delta_s$, the rational solution space of (7.2) has dimension $\max(s - 3, 0)$. In particular, a nonzero integer solution requires at least four scales, and every set of four distinct scales admits one. For four scales all solutions are multiples of*

$$r_i = K \frac{\delta_i}{\prod_{j \neq i} (\delta_i - \delta_j)}, \quad i = 1, 2, 3, 4. \quad (7.3)$$

The scalar K may be chosen to make all four exponents integers and then primitive by dividing their common divisor.

Proof. Put $x_i = r_i / \delta_i$. The three rows become $\sum x_i \delta_i^j = 0$ for $j = 0, 1, 2$. The resulting $3 \times s$ Vandermonde matrix has rank $\min(3, s)$: every square minor of size at most three on distinct scales is nonzero. This proves the dimension and the lower bound. For four scales, the leading-coefficient identity in Lagrange polynomial interpolation gives

$$\sum_{i=1}^4 \frac{\delta_i^j}{\prod_{l \neq i} (\delta_i - \delta_l)} = 0, \quad j = 0, 1, 2.$$

It proves (7.3); the one-dimensional kernel proves exhaustiveness. The entries are nonzero rational numbers, so clearing denominators gives the claimed integer vector. $\square$

This minimum concerns distinct Euler factors at $q = 1$, not arbitrary special-function representations or every other cusp. At other cusp denominators the first row in (7.1) changes and its rank must be checked afresh.

7.2 Intrinsic ramification and unbounded complexity with four factors

An arbitrarily chosen common multiple N can make the number m in Theorem 5.1 artificially large. The following criterion separates this coordinate effect from actual ramification of the *denominator correction*.

Theorem 7.3 (Exact correction ramification). *Let $H(Q) = 1 + h_m Q^m + \dots$, $h_m \neq 0$, be a nonconstant holomorphic germ, and put*

$$g = \gcd\{n \geq 1 : [Q^n](H(Q) - 1) \neq 0\}.$$

Then $g \mid m$. In the flat inverse with $u = (Y - C)/(Ch_m)$, the correction $V(u^{1/m})$ of Theorem 5.1 has minimal finite ramification degree m/g . The logarithmic term still has infinite monodromy and is not included in this finite-degree assertion.

For a flat Euler quotient at $q = 1$, take N to be the least common multiple of its active scales. Then $g = 1$ and this degree is exactly

$$\frac{N}{\delta_{\max}}. \tag{7.4}$$

Every positive integer occurs as this degree for a flat quotient with exactly four distinct Euler factors. Four factors are minimal.

Proof. For an m th root of unity ω , the condition $V(\omega s) = V(s)$ is equivalent, by $V(0) = 0$, to $\psi(\omega s) = \omega \psi(s)$. Since ψ is the inverse of $f(Q) = Q((H(Q) - 1)/(h_m Q^m))^{1/m}$, this is equivalent to $f(\omega Q) = \omega f(Q)$, and hence to $H(\omega Q) = H(Q)$. The last equivalence in the reverse direction uses the unique root with constant term 1. The rotation stabilizer of H inside the m th roots of unity is exactly the group of g th roots of unity: its coefficients require $\omega^n = 1$ for every occupied exponent. Thus exactly m/g circuits in u are needed to restore the correction. Equivalently its series lies in $\mathbb{C}\{u^{g/m}\}$ and in no smaller finite ramification. This includes $V = 0$, for which $H - 1$ is a single monomial and $g = m$.

For completeness, for any nontrivial finite Euler quotient $H(Q) = \prod_e \mathcal{P}(Q^e)^{r_e}$ the gcd of the support of $H - 1$ equals the gcd of its active e . One divisibility is immediate by writing every factor as a series in $Q^{\gcd e}$. For the converse, let d divide the entire support of $H - 1$. Then $\log H$ also has support in $d\mathbb{N}$. If some active e is not divisible by d , choose the smallest such e . Every smaller active scale that divides e would have to be divisible by d , which is impossible. The coefficient of Q^e in $\log H$ is therefore $-r_e \neq 0$, a contradiction. At $q = 1$ the active scales of H are $e = N/\delta$. Their gcd is 1 when N is the least common multiple, and (6.12) gives (7.4).

For every integer $m \geq 3$, choose the four scales $1, 2, m, m + 1$. They are distinct, their least common multiple is $m(m + 1)$, and their maximum is $m + 1$. Choose the nonzero integer kernel vector given by Theorem 7.2. It has all four entries nonzero, is flat, and has intrinsic correction degree m . Degree 1 is realized by scales $1, 2, 3, 6$, and degree 2 by $2, 3, 4, 6$, again with their nonzero kernel vectors. Minimality follows from Theorem 7.2. $\square$

This yields unbounded branch complexity with a fixed number of product factors. It also shows precisely which notion of ramification is meant: finite ramification of the analytic phase correction, superimposed on an already infinitely sheeted logarithmic inverse.

7.3 Arithmetic action ratios in the essential regime

There is a complementary restriction when $A_k \neq 0$. Set

$$M_k = \sum_{\delta|N} r_\delta(N/\delta) \gcd(k, \delta)^2 \in \mathbb{Z}.$$

Then $A_k = \pi^2 M_k / (6k^2 N)$. After replacing the target by its reciprocal when $M_k < 0$, the ratio between the dual action and the dominant action is

$$\lambda = \frac{a}{|A_k|} = \frac{24}{|M_k|} \in \mathbb{Q}_{>0}. \quad (7.5)$$

With $\epsilon = \text{sign}(M_k)$, $\beta' = \epsilon\beta$, $C' = C_{h,k}^\epsilon$, and target $Y = \mathcal{F}_r^\epsilon$, equation (3.10) gives

$$Q^{|M_k|} = \left(\frac{Y}{C'|A_k|^{\beta'}} \right)^{24} x_0^{24\beta'}, \quad 24\beta' = -12\epsilon R_0 \in \mathbb{Z}. \quad (7.6)$$

Thus the small dual parameter is algebraic over the pair (Y, x_0) ; all its fractional powers can be handled on a finite algebraic cover. This does not eliminate the logarithmic core or assert that the full inverse is algebraic. It does rule out irrational action-ratio ramification for this finite integer Euler-product family, in contrast to Proposition 3.2 for general prepared transseries.

8 Two explicit infinitely flat q -analogs

8.1 A four-factor quotient with an unramified correction

Consider

$$F_6(q) = \frac{\mathcal{P}(q^2)^5 \mathcal{P}(q^6)}{\mathcal{P}(q) \mathcal{P}(q^3)^5}, \quad Y(t) = \frac{8}{9} F_6(e^{-t}). \quad (8.1)$$

Its exponents on 1, 2, 3, 6 are $(-1, 5, -5, 1)$ and satisfy (7.2). The cusp constant is $C_{0,1} = 9/8$. With

$$a = \frac{2\pi^2}{3}, \quad Q = e^{-a/t},$$

Theorem 6.1 gives the exact identity

$$Y(t) = H(Q) = \frac{\mathcal{P}(Q) \mathcal{P}(Q^3)^5}{\mathcal{P}(Q^2)^5 \mathcal{P}(Q^6)}. \quad (8.2)$$

Equivalently, $F_6(e^{-t}) F_6(e^{-a/t}) = 9/8$. This self-reciprocity is a consequence of the classical eta law, not a new modular transformation.

The first terms, computed by (6.10)–(6.11), are

$$\begin{aligned} H(Q) &= 1 - Q + 4Q^2 - 10Q^3 + 20Q^4 - 39Q^5 \\ &\quad + 76Q^6 - 140Q^7 + 244Q^8 - 415Q^9 + O(Q^{10}). \end{aligned} \quad (8.3)$$

Put $u = 1 - Y$. The inverse $Q = \psi(u)$ and its logarithmic unit are

$$\begin{aligned} \psi(u) &= u + 4u^2 + 22u^3 + 140u^4 + 965u^5 + 6992u^6 \\ &\quad + 52420u^7 + 402880u^8 + 3155410u^9 + 25083336u^{10} + O(u^{11}), \end{aligned} \quad (8.4)$$

$$V(u) = 4u + 14u^2 + \frac{220}{3}u^3 + 451u^4 + \frac{15124}{5}u^5 + \frac{64274}{3}u^6 + O(u^7). \quad (8.5)$$

All these series converge near zero. The complete inverse family is

$$t_\ell(Y) = -\frac{2\pi^2/3}{\text{Log}(1-Y) + 2\pi i\ell + V(1-Y)}, \quad \ell \in \mathbb{Z}. \quad (8.6)$$

The real small-positive- t inverse uses $u > 0$ and $\ell = 0$. The numerator $2\pi^2/3$ and the full analytic denominator correction are invisible in the algebraic expansion $Y \sim 1$.

For the same quotient at the four divisor cusp denominators, the normalized actions are

$$\left(\frac{6k^2}{\pi^2} A_k \right)_{k=1,2,3,6} = (0, 8, -12, 0). \quad (8.7)$$

Thus the cusps of denominators 1 and 6 are flat, the denominator-2 cusp has $A_2 = \pi^2/3 > 0$, and the denominator-3 cusp has $A_3 = -2\pi^2/9 < 0$. At $\zeta_6 = e^{\pi i/3}$, the constant is $C_{1,6} = 1$ and the first transformed coefficient gives

$$F_6(\zeta_6 e^{-t}) = 1 + e^{-\pi i/3} \exp\left(-\frac{\pi^2}{9t}\right) + o\left(\exp\left(-\frac{\pi^2}{9t}\right)\right) \quad (8.8)$$

along positive t . Here $a = \pi^2/54$ and the first nonzero index is $m = 6$. In fact all transformed exponents are multiples of 6, so the primitive coordinate is $e^{-\pi^2/(9t)}$. This illustrates why an auxiliary Puiseux order must not automatically be called intrinsic.

8.2 A four-factor quotient with genuine square-root correction

Consider the distinct scales 2, 3, 4, 6 and exponents $(-1, 4, -4, 1)$:

$$F_{12}(q) = \frac{\mathcal{P}(q^3)^4 \mathcal{P}(q^6)}{\mathcal{P}(q^2) \mathcal{P}(q^4)^4}, \quad C = \frac{16\sqrt{3}}{27}, \quad Y(t) = F_{12}(e^{-t})/C. \quad (8.9)$$

Here $N = 12$, $a = \pi^2/3$, and

$$Y(t) = H(Q) = \frac{\mathcal{P}(Q^2) \mathcal{P}(Q^4)^4}{\mathcal{P}(Q^3)^4 \mathcal{P}(Q^6)}, \quad Q = e^{-a/t}. \quad (8.10)$$

The smallest powers are 2 and 3, so the primitive action coordinate cannot remove their relative square-root correction. Specifically,

$$H(Q) = 1 - Q^2 + 4Q^3 - 5Q^4 - 4Q^5 + 19Q^6 - 20Q^7 - 9Q^8 + 60Q^9 - 76Q^{10} + O(Q^{11}). \quad (8.11)$$

Put $u = 1 - Y$ and fix $s = \sqrt{u}$ coherently with $\log s = \frac{1}{2} \text{Log } u$. The analytic inverse of $s = Q\sqrt{(1-H(Q))/Q^2}$ is

$$\begin{aligned} \psi(s) = s + 2s^2 + \frac{15}{2}s^3 + 32s^4 + \frac{1203}{8}s^5 + 746s^6 \\ + \frac{61467}{16}s^7 + 20320s^8 + \frac{14038275}{128}s^9 + 601374s^{10} + O(s^{11}), \end{aligned} \quad (8.12)$$

$$V(s) = 2s + \frac{11}{2}s^2 + \frac{59}{3}s^3 + \frac{337}{4}s^4 + \frac{7843}{20}s^5 + \frac{11567}{6}s^6 + O(s^7). \quad (8.13)$$

All inverse sheets are

$$t_\ell(Y) = -\frac{\pi^2/3}{\frac{1}{2} \text{Log}(1-Y) + \pi i\ell + V((-1)^\ell \sqrt{1-Y})}, \quad \ell \in \mathbb{Z}. \quad (8.14)$$

One circuit of Y around 1 changes the sign of the square root and increments the sheet. Two circuits restore the square-root sign but add $2\pi i$ to the logarithm. Treating the square root and logarithm as independent principal branches would lose this compatibility.

Both examples have leading real inverse $t \sim -2\pi^2/(3 \operatorname{Log}(1-Y))$. Their leading logarithmic scale is therefore identical, but one denominator correction starts at $1-Y$ and the other at $\sqrt{1-Y}$. A leading-equivalent inverse does not determine the subdominant ramification.

9 Recovering the quotient from its cusp actions

An individual cusp may hide every algebraic coefficient, and some cusps may have zero essential action. Nevertheless, the actions at the *divisor cusps together* determine the original finite exponent vector.

For every $k \mid N$, set

$$a_k = \frac{6k^2}{\pi^2} A_k = \sum_{\delta \mid N} \frac{r_\delta}{\delta} \gcd(k, \delta)^2. \quad (9.1)$$

Let μ denote the Möbius function and

$$J_2(d) = \sum_{e \mid d} \mu(d/e) e^2 = d^2 \prod_{p \mid d} (1 - p^{-2}) > 0$$

be the second Jordan totient.

Theorem 9.1 (Double Möbius recovery). *For every $\delta \mid N$,*

$$r_\delta = \delta \sum_{\substack{m \mid N \\ \delta \mid m}} \frac{\mu(m/\delta)}{J_2(m)} \sum_{e \mid m} \mu(m/e) a_e. \quad (9.2)$$

The linear map from $(r_\delta)_{\delta \mid N}$ to $(a_k)_{k \mid N}$ is invertible, with determinant

$$\prod_{d \mid N} \frac{J_2(d)}{d} \quad (9.3)$$

when rows and columns use the same increasing divisor order. Consequently a nontrivial Euler quotient cannot have zero essential action at every divisor cusp, and in particular cannot be infinitely flat at all of them.

Proof. The divisor identity $n^2 = \sum_{d \mid n} J_2(d)$ gives

$$a_k = \sum_{d \mid k} J_2(d) \sum_{\substack{\delta \mid N \\ d \mid \delta}} \frac{r_\delta}{\delta}.$$

Möbius inversion on the divisors of k therefore gives

$$b_d := \frac{1}{J_2(d)} \sum_{e \mid d} \mu(d/e) a_e = \sum_{\substack{\delta \mid N \\ d \mid \delta}} \frac{r_\delta}{\delta}.$$

Apply Möbius inversion a second time, now to this upper-divisor sum: $r_\delta/\delta = \sum_{\delta \mid m \mid N} \mu(m/\delta) b_m$. This is (9.2).

For the determinant, let $E_{k,d} = 1$ when $d \mid k$ and 0 otherwise. With divisors increasing, E is triangular with diagonal 1. The matrix in (9.1) factors as

$$E \operatorname{diag}(J_2(d)) E^\top \operatorname{diag}(1/\delta).$$

Taking determinants proves (9.3). If all a_k vanish, recovery gives $\mathbf{r} = 0$, proving the final statements. $\square$

This theorem does not assume modular-form invariance of the quotient. It is a finite arithmetic consequence of the exact cusp actions. The underlying divisor-matrix factorization is elementary; the contribution here is its explicit use as an inverse-data reconstruction procedure, not a claim to have discovered Möbius inversion or modular rigidity.

The actions can themselves be read from leading radial inverse data. Indeed, on a continuously chosen logarithm along the radial image,

$$\lim_{t \downarrow 0} t \log \mathcal{F}_{\mathbf{r}}(\zeta e^{-t}) = -A_k. \quad (9.4)$$

Equivalently, along the corresponding inverse branch, $t(Y) \log Y \rightarrow -A_k$. The real version $t \log |\mathcal{F}_{\mathbf{r}}(\zeta e^{-t})| \rightarrow -A_k$ avoids the logarithmic phase convention. Even in the zero-action cases the limit exists and is zero. Exact action data at all $k \mid N$ therefore recover every exponent. No analogous recovery from a *single* flat cusp's leading constant is asserted. For F_6 , substituting (8.7) into (9.2) returns $(-1, 5, -5, 1)$.

10 A nonmodular application: fixed-parameter shifted factorials

The exact modular tail is an advantage of the Euler-product family, not an assumption that should silently be transferred to every q -analog. We now give an all-orders inverse in a case where the algebraic series need not converge. This makes the formal/analytic distinction concrete.

Fix $z \in (0, 1)$ and define

$$P_z(t) = (z; e^{-t})_\infty = \prod_{n \geq 0} (1 - z e^{-nt}), \quad \Re t > 0.$$

The following constants are positive real when appropriate:

$$A = \operatorname{Li}_2(z) > 0, \quad C = \sqrt{1-z} > 0, \quad b_j = \frac{B_{2j}}{(2j)!} \operatorname{Li}_{2-2j}(z).$$

Here B_n are the Bernoulli numbers with $B_2 = 1/6$, $B_4 = -1/30$, and $\operatorname{Li}_s(z) = \sum_{n \geq 1} z^n/n^s$ for $|z| < 1$.

Proposition 10.1 (All-orders inverse of a shifted factorial). *Uniformly for z in a compact subinterval of $(0, 1)$ and $t \rightarrow 0$ in $|\arg t| \leq \pi/2 - \epsilon$, for every fixed $M \geq 0$,*

$$\log P_z(t) = -\frac{A}{t} + \log C - \sum_{j=1}^M b_j t^{2j-1} + O(t^{2M+1}). \quad (10.1)$$

For $Y = P_z(t)$, choose $L = \operatorname{Log}(C/Y)$ on the corresponding logarithmic cover. The local inverse has the all-orders expansion

$$t \sim \sum_{n \geq 1} \frac{L^{-n}}{n} [w^{n-1}] \left(A + \sum_{j \geq 1} b_j w^{2j} \right)^n. \quad (10.2)$$

Every coefficient involves only finitely many b_j , and only odd powers of L^{-1} occur. In particular,

$$t = \frac{A}{L} + \frac{A^2 \operatorname{Li}_0(z)}{12L^3} + \frac{1}{L^5} \left(\frac{A^3 \operatorname{Li}_0(z)^2}{72} - \frac{A^4 \operatorname{Li}_{-2}(z)}{720} \right) + O(L^{-7}). \quad (10.3)$$

These are asymptotic assertions, not a claim of convergence of (10.2) or of a specified Borel sum.

Proof. The absolutely convergent logarithmic expansion gives

$$\log P_z(t) = - \sum_{n \geq 1} \frac{z^n}{n(1 - e^{-nt})}. \quad (10.4)$$

On a fixed closed sector $|\arg w| \leq \pi/2 - \epsilon$, the kernel has

$$\frac{1}{1 - e^{-w}} = \frac{1}{w} + \frac{1}{2} + \sum_{j=1}^M \frac{B_{2j}}{(2j)!} w^{2j-1} + R_M(w), \quad |R_M(w)| \leq C_{M,\epsilon} |w|^{2M+1}.$$

Near zero this follows from its convergent Taylor expansion after removing the pole. For $|w| \geq 1$, the denominator is bounded away from zero because $\Re w \geq |w| \sin \epsilon$, and the same bound follows by bounding the finitely many subtracted terms. Inserting $w = nt$ into (10.4) bounds the remainder by $C_{M,\epsilon} |t|^{2M+1} \sum_{n \geq 1} z^n n^{2M}$, uniformly on the stated compact sets. Summing the individual terms proves (10.1).

Formally, the logarithmic equation is $L = A/t + \sum_{j \geq 1} b_j t^{2j-1}$, or

$$t = L^{-1} \left(A + \sum_{j \geq 1} b_j t^{2j} \right).$$

Lagrange inversion proves (10.2) in the formal power-series ring. Parity of the parenthesis proves the odd-power claim. The first coefficients are A , $b_1 A^2$, and $2b_1^2 A^3 + b_2 A^4$, which give (10.3).

To identify this formal inverse with an analytic asymptotic inverse, put $x = A/t$. The exact forward logarithmic phase is $x + O(x^{-1})$, with derivative $1 + O(x^{-2})$ on smaller large- x sectors. The derivative estimate follows either by differentiating the normally convergent logarithmic sum on interior sectors or by Cauchy's estimate applied to the uniform remainder on a slightly larger sector. For large L , Lemma 2.2 on a disk about L of radius $K/|L|$, with K sufficiently large, gives the unique root close to L . At every fixed order, substitution of the formal inverse cancels the corresponding terms of (10.1); the same certificate transports the remaining phase error into an inverse error. Taking arbitrarily large fixed M proves the full asymptotic statement. $\square$

The forward expansion is part of the classical polylogarithm–Bernoulli q -factorial calculus [7]; we included its proof to make the inverse statement and its sectorial remainder independent of a formal citation. The recent gamma-product approach of Arabi Ardehali and Rosengren provides a broader route to nonmodular asymptotics [10]. At $z \rightarrow 1$ our constants and remainder bounds are not uniform: $C \rightarrow 0$ and the negative-index polylogarithms diverge. A simultaneous $z = e^{-\alpha t} \rightarrow 1$ limit requires a different, typically gamma-based, core. Substituting $z = 1$ into (10.3) is invalid.

11 Finite Gaussian multinomials: ordinary critical-point reversal

For comparison with the infinitely flat boundary case, let $n_1, \dots, n_s$ be nonnegative integers with sum n , and consider the Gaussian multinomial polynomial

$$P(q) = \left[\begin{matrix} n \\ n_1, \dots, n_s \end{matrix} \right]_q = \frac{\prod_{j=1}^n (1 - q^j)}{\prod_{i=1}^s \prod_{j=1}^{n_i} (1 - q^j)}. \quad (11.1)$$

These are finite q -analogs; the standard q -binomial notation and product identities are recorded in DLMF §17.2 [6].

Proposition 11.1 (Root-of-unity order and inverse sheets). *At a primitive k th root ζ , the vanishing order of P is exactly*

$$\nu = \left\lfloor \frac{n}{k} \right\rfloor - \sum_{i=1}^s \left\lfloor \frac{n_i}{k} \right\rfloor, \quad 0 \leq \nu \leq s - 1. \quad (11.2)$$

If $\nu > 0$, write $P(\zeta + w) = cw^\nu U(w)$ with $c \neq 0$ and $U(0) = 1$. There are ν ordinary Puiseux inverse sheets, obtained by evaluating

$$w(z) = \sum_{j \geq 1} \frac{z^j}{j} [w^{j-1}] U(w)^{-j/\nu}, \quad z = (Y/c)^{1/\nu}, \quad (11.3)$$

at the ν choices of the root. The series is convergent near zero.

Proof. Each factor $1 - q^j$ vanishes simply at ζ precisely when $k \mid j$. Counting numerator and denominator factors proves the order formula; all remaining local units have nonzero values, so no further cancellation can change it. Write $n_i = ka_i + b_i$ with $0 \leq b_i < k$; then $\nu = \lfloor \sum_i b_i/k \rfloor$, proving the bounds. For $\nu > 0$, the analytic function $wU(w)^{1/\nu}$ has derivative 1 at zero. Its inverse exists and Lagrange inversion gives (11.3). $\square$

If $\nu = 0$, the polynomial has a nonzero value at the cusp, but its derivative need not be nonzero. To study its inverse at that value one must find the first nonzero derivative of $P(q) - P(\zeta)$; the floor formula alone does not determine that critical order.

For example,

$$\left[\begin{matrix} 6 \\ 2, 2, 2 \end{matrix} \right]_q = \Phi_3(q)^2 \Phi_4(q) \Phi_5(q) \Phi_6(q), \quad (11.4)$$

where Φ_j is the j th cyclotomic polynomial. At $\zeta = e^{2\pi i/3}$ this equals $6\zeta(q - \zeta)^2 + O((q - \zeta)^3)$, and the inverse branches are

$$q = \zeta \pm \sqrt{Y/(6\zeta)} + O(Y).$$

Here the critical point has a finite-order zero and ordinary Puiseux reversion suffices. In Section 8, by contrast, the boundary limit has no first nonzero algebraic derivative in the cusp coordinate. Its logarithmic denominator is essential, not an optional re-expression of an ordinary Taylor inverse.

12 Sectorial transitions and exponentially small inverse jumps

Local formulas must be glued with care. Let F_- and F_+ be holomorphic and univalent on chosen overlapping source charts, with local inverses G_- and G_+ . If a source transition S satisfies $F_+ = F_- \circ S$, then on the corresponding target overlap

$$G_+ = S^{-1} \circ G_- . \quad (12.1)$$

If instead the transition acts on the target, $F_+ = T \circ F_-$, then

$$G_+ = G_- \circ T^{-1} . \quad (12.2)$$

These exact identities follow by composition and are not interchangeable. They apply to actual sectorial sums when the necessary summability and univalence hypotheses have been established. They do not themselves prove resurgence or construct a Stokes automorphism.

Proposition 12.1 (All-order perturbation transport). *Let F be univalent with inverse G , and let Δ be holomorphic on a source neighborhood. For sufficiently small ϵ , the inverse of $F_\epsilon = F + \epsilon\Delta$ satisfies locally*

$$G_\epsilon(y) = G(y) + \sum_{n \geq 1} \frac{(-\epsilon)^n}{n!} \left(\frac{d}{dy} \right)^{n-1} [G'(y) \Delta(G(y))^n] . \quad (12.3)$$

The expansion converges on sufficiently small compact subcharts for small ϵ . In particular,

$$\left. \frac{\partial G_\epsilon(y)}{\partial \epsilon} \right|_0 = - \frac{\Delta(G(y))}{F'(G(y))} . \quad (12.4)$$

Proof. Put $u = F(z)$ and $d(u) = \Delta(G(u))$. Then $y = u + \epsilon d(u)$ and $z = G(u)$. Lagrange inversion with the observable G gives (12.3). The analytic implicit-function theorem and a sufficiently small ϵ disk justify convergence uniformly on compact subcharts. The $n = 1$ term is (12.4); it also follows by differentiating $F_\epsilon(G_\epsilon(y)) = y$. $\square$

Sauzin's nonlinear resurgent analysis supplies rigorous settings in which nonlinear substitutions and inversions preserve appropriate resurgent and summability classes [4]. Applying those results to a particular divergent transseries requires checking its precise continuability and growth hypotheses. In the exact modular examples, H and its inverse germ already converge: no unspecified Borel sum is required to justify their local formulas. Analytic monodromy of their logarithms must still be kept distinct from changes between lateral sums of a divergent expansion.

12.1 An inverse action budget at a flat limit

The first-order jump law displays a useful threshold. Suppose, along a closed decay sector,

$$F(t) = Y_* + ce^{-A_0/t}(1 + O(e^{-a/t})), \quad \Delta(t) = be^{-B_0/t},$$

where $A_0, a, B_0 > 0$ and $b, c \neq 0$. Then

$$\left. \partial_\epsilon G_\epsilon(F(t)) \right|_0 \sim - \frac{bt^2}{cA_0} e^{-(B_0 - A_0)/t} . \quad (12.5)$$

This follows by differentiating the leading forward term and applying (12.4). Forward action B_0 becomes inverse action $B_0 - A_0$, dressed by t^2 . It is therefore incorrect to transfer the forward error exponent unchanged through an infinitely flat inverse.

When $B_0 > A_0$, a bounded perturbation coefficient has an exponentially small ratio to the leading displacement; a small inverse-phase chart remains appropriate. When $B_0 = A_0$, it changes the leading amplitude c to $c + \epsilon b$ (unless that coefficient cancels, in which case a new leading action is needed). When $0 < B_0 < A_0$, any fixed nonzero ϵb eventually changes the leading action itself. The linear-response formula is still a derivative at $\epsilon = 0$, but is not a uniform approximation for fixed ϵ in that last regime. These distinctions specify an inverse error budget without conflating a small absolute forward error with a small relative error in the quantity that is being inverted.

13 Algorithms, verification, and reproducibility

13.1 A branch-aware workflow

The preceding proofs suggest a practical algorithm rather than only existence assertions. Given an Euler quotient, a cusp, a target, and a chosen continuation sheet, first compute the integer and cyclotomic data in (6.2)–(6.7). Evaluate the three invariants exactly and select the chart in Theorem 7.1. Next generate the required coefficients using (6.10) and (6.11). In the flat chart find the first action, remove any redundant primitive-coordinate factor, and compute the analytic inverse ψ and V with (5.3). The logarithm and root are continued together, as in (5.2); a principal logarithm is not a universal sheet-selection algorithm.

Finally evaluate a truncated inverse and certify its chart whenever explicit analytic bounds are available. Corollary 5.2 provides a closed tail estimate from R, M ; in more general charts Lemma 2.2 converts a computable residual and a local derivative bound into existence and error control. The existence of analytic radii in a theorem is not by itself a numerical interval certificate for an implementation. Deriving sharp effective radii is a separate, useful task.

For exact coefficient computation, the following recurrence avoids symbolic differentiation. If $A(z) = \sum a_n z^n$, $a_0 = 1$, and $B(z) = A(z)^p = \sum b_n z^n$, then $b_0 = 1$ and

$$nb_n = \sum_{k=1}^n ((p+1)k - n) a_k b_{n-k}. \quad (13.1)$$

It follows by comparing coefficients in $AB' = pA'B$. Rational p and rational input coefficients therefore remain exact. The supplied program uses rational arithmetic for this recurrence, Lagrange coefficients, compositions, and cusp recovery.

13.2 Numerical product tails

For $|q| = r < 1$, truncating the logarithm of $\mathcal{P}(q)$ after factor M has the absolute tail bound

$$\left| \sum_{j>M} \log(1 - q^j) \right| \leq \frac{r^{M+1}}{(1-r)(1-r^{M+1})}. \quad (13.2)$$

Indeed, $|\log(1 - q^j)| \leq \sum_{l \geq 1} r^{jl}/l \leq r^j/(1 - r^j)$, and summing the geometric bound proves the formula. For quotients, multiply the individual bounds by the absolute values of their integer exponents and add. The program chooses a cutoff using this type of bound but evaluates the finite sum in ordinary arbitrary precision, not with outward-rounded arithmetic. Its recorded numerical agreement is therefore a diagnostic, not a machine-certified enclosure.

13.3 Executed checks

The package contains `code/verify.py`, which has no network dependency and uses Python's rational arithmetic and `mpmath`. The recorded run checks exact reversion compositions for both examples, all three flatness moments, the four-node kernel formula, and recovery on every basis vector for levels 1, 2, 6, 12, 24, 30, 60. It also checks the arbitrary-degree four-factor construction, the displayed shifted-factorial reversion coefficients on exact rational test inputs, and the finite Gaussian example. The recorded run passed 619 exact assertions; software versions and individual coefficient data are recorded in `data/verification.json`.

Fifteen independent forward-versus-dual product tests use three exponent vectors, including one with nonzero weight and essential action, at cusps $(h, k) = (0, 1), (1, 2), (1, 3), (2, 5), (1, 6)$. All use $t = 0.37 + 0.09i$ at 115 decimal digits, and all relative discrepancies are below 10^{-90} . This tests, in particular, the Dedekind phases in (6.9), not only the positive-real $q \rightarrow 1$ case.

Table 1 evaluates the first example from its *original* product at $q = e^{-t}$ and reconstructs t from the inverse polynomial $\psi_N(u)$, using $-a/(\log \psi_N(u) + 2\pi i \ell)$. Here N is the degree of ψ_N , not of V_N . Truncating ψ at degree N generally gives a logarithmic-unit error $O(u^N)$, so the indexing differs by one from Corollary 5.2. The complex row specifically requires a

Reference t	Sheet	$N = 1$ error	$N = 4$ error	$N = 8$ error
0.5	0	$2.93 \cdot 10^{-7}$	$5.06 \cdot 10^{-22}$	$2.28 \cdot 10^{-41}$
0.35	0	$5.10 \cdot 10^{-10}$	$3.95 \cdot 10^{-32}$	$2.84 \cdot 10^{-61}$
0.25	0	$1.41 \cdot 10^{-13}$	$1.74 \cdot 10^{-45}$	$1.08 \cdot 10^{-87}$
$0.4 + 0.08i$	1	$1.37 \cdot 10^{-8}$	$8.15 \cdot 10^{-27}$	$8.90 \cdot 10^{-51}$

Table 1: Absolute inverse errors from the executed non-interval checks. The sheet is selected to match the known reference point, not inferred from a principal-logarithm convention.

nonprincipal logarithmic sheet. Using the principal logarithm would converge to a different inverse, not to the displayed reference point. The test computes the matching integer from the known reference phase. An actual inverse application must supply its sheet by initial data or continuation rather than use an unavailable reference solution.

Two additional checks use $s = \pm 10^{-5}$ in the second example. With ψ through degree ten, the forward residuals in $1 - H(\psi(s)) - s^2$ are below $7 \cdot 10^{-54}$. These test the two distinct square-root branches in the convergent dual germ. Neither these tests nor the exact finite computations constitute a proof of an untested infinite theorem; the conventional proofs above provide that part of the argument.

14 Further research questions and topics

The following program distinguishes extensions with concrete technical obstacles from claims already proved here.

R1. Countable actions and accumulation. Find useful complete topological algebras in which the parameter-lift argument survives countably many actions with finite positive-projection budgets. Determine what replaces this budget when actions accumulate at a positive value or at zero. The formal contraction lemma is available, but admissibility of substitution and analytic realization are not automatic.

R2. Divergent analytic units and summation-compatible inversion. Replace the convergent unit U in Theorem 3.1 by a specified resurgent or multisummable unit.

Prove an inverse theorem with explicit lateral-sum compatibility, and compute which singularities appear in the Borel plane after the dominant-phase change. Established closure theorems are starting points, not substitutes for their hypothesis checks in each q -family.

- R3. Confluence of essential and flat charts.** For continuously weighted products or shifted parameters, let the essential action tend to zero simultaneously with t . Construct a uniform chart that connects alternatives (I), (II), (III), and (IV) of Theorem 7.1. A fixed nonzero-action inverse is singular in this transition, and merely setting $A_k = 0$ in its Lambert core cannot be justified.
- R4. Effective first-action bounds at arbitrary cusps.** Lemma 6.2 proves termination of a coefficient search, but not a sharp complexity bound. Bound the first nonzero index and the gcd of the occupied support in terms of N, k and the active exponent vector, including cancellations among cyclotomic phases. Identify a closed form analogous to (6.12) when one exists.
- R5. Minimum support away from $q = 1$.** Classify the rank drops of the three rows in (7.1) at arbitrary k . Determine exactly which cusps admit two- or three-factor flat quotients, and classify their intrinsic correction ramification. The four-factor result here is specific to $q = 1$.
- R6. Effective inverse radii and rigorous enclosures.** For concrete H , locate the nearest critical values of $Q((H(Q) - 1)/(h_m Q^m))^{1/m}$ and derive numerical R, M for Corollary 5.2. Build outward-rounded complex interval certificates for both sheet continuation and inverse truncation. The existing numerical code is a starting test suite, not such a certificate.
- R7. Optimal truncation with intact action blocks.** When algebraic blocks diverge but the action expansion converges, choose joint algebraic and action cutoffs that meet a prescribed inverse error. The action subtraction in (12.5) shows why a forward optimal cutoff can be suboptimal or insufficient after inversion. A useful theorem should distinguish fixed-action accuracy from total absolute accuracy.
- R8. Moving shifted parameters and gamma cores.** Study $(z; q)_\infty$ with $z = e^{-\alpha t}$, including exceptional α near poles and zeros of the limiting gamma factors. Combine the recent product representation [10] with a uniform inverse chart that remains valid as $z \rightarrow 1$. Include root-of-unity analogues and coalescing gamma singularities.
- R9. Growing cusp denominator.** The results here keep k, N fixed. Determine a useful joint limit in which $k \rightarrow \infty$ while $t \rightarrow 0$ and the dual parameter $\exp(-4\pi^2/(k^2 N t))$ remains small. Uniform control must also track the phases, the first occupied dual index, and inverse critical values; density of rational cusps prevents a naive global patching argument.
- R10. Stable recovery from noisy cusp data.** Analyze the condition numbers of the divisor-action matrix in Theorem 9.1. Exploit its prime-power tensor structure and integer exponent constraints to reconstruct quotients from approximate inverse measurements. Determine how many selected cusps suffice when the active support is sparse or only bounded a priori.
- R11. Ramification spectra with bounded exponent height.** Four factors suffice for every correction degree, but the integer exponents supplied by the Vandermonde kernel may be large. Determine sharp tradeoffs between ramification degree, maximum scale,

coefficient height, and number of factors. Classify symmetries that force correction coefficients to vanish without changing the leading action.

- R12. Basic hypergeometric functions and multiple saddles.** Extend the cusp-to-inverse strategy beyond products to basic hypergeometric series and Nahm-type sums. Multiple competing saddles introduce interference and possible derivative cancellation; root-of-unity asymptotics such as those in [8] provide forward input, but inverse branch selection must be proved independently.
- R13. Global transition data.** Combine logarithmic monodromy, finite correction ramification, and sectorial Stokes maps in a single explicitly based system of source and target charts. Determine when the local transitions in (12.1) satisfy a global compatibility law. Do not identify a dominance wall with a Stokes discontinuity without specifying an analytic summation problem.
- R14. Formalization in stages.** A practical Lean program could first formalize the finite divisor-matrix recovery, Vandermonde kernel, and rational coefficient recurrences; then the complete filtered fixed-point lemma and finite power-series identities; and finally the analytic inverse chart, error certificate, and modular bridge. The repository's existing exact modules and their stated limitations should be respected. No Lean formalization of the new theorems is claimed in this package.

15 Conclusion

A complex transseries inverse is not only a list of coefficients. It comprises a dominant phase, an admissible analytic or formal algebra, a collection of small actions, and coherent continuation data. For prepared exponential maps these ingredients lead to a convergent multivariable inverse. When the dominant essential action vanishes, slope-adapted and logarithmic–Puiseux charts provide the missing alternatives.

The q -product application makes this principle explicit. At every rational cusp, a finite Euler quotient has an exact elementary factor times a convergent dual product. Three computable invariants determine its inverse regime. In the infinitely flat regime, an ordinary analytic inverse in the dual variable becomes a convergent correction inside an infinitely sheeted logarithmic denominator. Four Euler factors are both necessary and sufficient at $q = 1$, and already realize every finite correction ramification degree. Despite local flatness, divisor-cusp actions collectively recover the full exponent vector.

These conclusions answer the scoped questions posed in Section 1, with proofs and reproducible checks. They do not establish a universal inversion theory for all transseries, a new modular transformation law, or worldwide priority. The remaining program is to extend the same explicit separation of algebra, asymptotics, and analytic continuation to divergent blocks, moving parameters, and multi-saddle q -analogs.

A Proof dependencies and research-status ledger

Result group	Main proof ingredients	Scope of the claim
Prepared inverse	Analytic implicit-function theorem; filtered contraction; Lagrange inversion	Holomorphic parameter unit and finite admissible action set; no general divergent-unit assertion.
Flat inverse atlas	Ordinary analytic inversion in Q ; logarithms on a cover	Exact local sheets and convergent correction with explicit Taylor-tail bound.
Cusp preparation	Classical eta transformation; finite products	Exact identity on $\Re t > 0$ for integer Euler quotients; not a new modular law.
Four regimes	Exact normal form; leading-term comparison	Necessary and sufficient flatness test at each fixed cusp.
Four-factor and ramification results	Vandermonde kernel; rotational stabilizers; Euler-product coefficient uniqueness	Minimum factor count at $q = 1$; exact correction degree; every degree attained with four factors.
Cusp-data recovery	Jordan divisor identity; two Möbius inversions	Exact finite-dimensional recovery with specified N and divisor-cusp measurements.
Shifted factorial	Kernel expansion; formal reversal; complex residual certificate	All finite asymptotic orders for fixed z in compact subsets of $(0, 1)$; no $z \rightarrow 1$ uniformity claim.
Finite multinomial	Count simple roots; analytic Puiseux inversion	Finite-order zeros at roots of unity; nonzero-value criticality treated separately.
Sectorial jump transport	Composition; Lagrange inversion with an observable	Exact chart identities and perturbation series; resurgent summation compatibility requires additional hypotheses.

The article supplies conventional proofs for these scoped statements. The established tools are attributed in the bibliography. The new formulations and combined classifications are proposed research results of this draft, not independently refereed or proof-assistant-checked claims. The repository comparison is restricted to the cited inspected documentation. The program's exact finite checks and floating-point diagnostics have the distinct evidential roles explained in Section 13.

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